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Independent Search and Vetting of TESS Objects of Interest Without Follow-up Observations Recorded on ExoFOP

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ABSTRACT

TESS Objects of Interest (TOIs) without follow-up observations recorded on ExoFOP offer a well-defined sample for reassessing follow-up priorities using archival photometry. We search the SPOC 120 s light curves of 132 stars hosting 133 such TOIs and recover 110 (83%) of the catalogued signals. The analysis combines periodic and isolated-transit searches with photometric, centroid and instrumental diagnostics, a convolutional neural network evaluated with cross-validation grouped by star, difference imaging, and `TRICERATOPS` false-positive probabilities. Four TOIs meet our candidate-selection criteria: TOI-3025.01, TOI-4674.01, TOI-7850.01 and TOI-7888.01. Three have significant difference images consistent with the target position; the fourth lacks a significant image. Their nearby false-positive probabilities are below 10^{-6}, but their false-positive probabilities are 1.7–34%, above the 1% validation threshold, because their radii (6.7–13.9 $R_{\oplus}$) do not exclude stellar companions. Supplementary tests identify two initially inconclusive TOIs with significant source offsets and four with source positions consistent with the target and false-positive probabilities of approximately 1–5%. We also report 15 signals that pass photometric vetting but have no match in the catalogues searched.

Keywords: Exoplanet detection methods (489) — Transit photometry (1709) — Convolutional neural networks (1938) — Exoplanet catalogs (488)

1. INTRODUCTION

The Transiting Exoplanet Survey Satellite (TESS; G. R. Ricker et al. 2015) has produced more than 8000 TESS Objects of Interest (TOIs; N. M. Guerrero et al. 2021). Their confirmation relies on the TESS Follow-up Observing Program (TFOP), which coordinates ground-based photometry, high-resolution imaging and spectroscopy. Follow-up capacity is limited, and on 2026 November 2 the ExoFOP archive listed 306 TOIs with a planet-candidate (PC) or empty working-group disposition and no recorded time-series, spectroscopic or imaging observations. Absence of an ExoFOP record does not establish that a target has never been observed elsewhere.

For such targets an independent, fully automated analysis of the public TESS data can help to prioritize follow-up by assessing signal recovery, testing for photometric signatures of eclipsing binaries, constraining the source position, and estimating probabilities for astrophysical false-positive scenarios. Machine-learning vetting of TESS signals (e.g. L. Yu et al. 2019; E. Tey et al. 2023; H. Valizadegan et al. 2022) and statistical validation with `TRICERATOPS` (S. Giacalone et al. 2021) are established tools; here they are combined with a blind search, a long-period single-transit search and a pixel-level source test in one pipeline, and applied to the open TOIs. ExoMiner++ (H. Valizadegan et al. 2025) already provides deep-learning classifications for TESS two-minute threshold-crossing events using several diagnostic inputs. Our contribution is a reassessment of the TOIs without recorded follow-up, with a blind search of their light curves and difference-image source tests.

Section 2 describes the data and the sample, Section 3 the pipeline and its calibration, Section 4 the results, and Section 5 their limitations.

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2. DATA AND SAMPLE

2.1. *Light curves*

We use the SPOC (J. M. Jenkins et al. 2016) 120 s light curves (PDCSAP flux, default quality mask) of every sector available on MAST, with the flux-weighted centroids (MOM_CENTR) and the aperture extension of each file. Stellar parameters come from the TIC v8 (K. G. Stassun et al. 2019).

2.2. *Open TOIs*

The ExoFOP TOI table was retrieved on 2026 November 2. We define a TOI as *open* when its TFOP working-group disposition is PC (or empty) and the numbers of time-series, spectroscopic and imaging observations recorded on ExoFOP are all zero. This yields 306 TOIs on 304 stars, of which 132 stars (133 TOIs) have SPOC 120 s light curves in our data set; the remaining targets are outside the scope of this work. The host stars have TESS magnitudes of 7.3–16.4 (median 13.1) and 1–39 (median 3) sectors of 120 s data.

2.3. *Training set*

The classifier is trained on TOIs with final dispositions: confirmed (CP) and known (KP) planets as positives, false positives (FP) and false alarms (FA) as negatives, using every 120 s sector of their host stars (2134 stars). PC and APC TOIs are not used for training.

3. METHODS

3.1. *Detrending*

Light curves are detrended with a Tukey biweight in a sliding time window (the method of `wotan`; M. Hippke et al. 2019), evaluated per data segment so that windows never span gaps. The blind search uses a 0.75 d window, chosen on a tuning set of injected transits. When a transit ephemeris is known, points in transit get zero weight and the window is three transit durations (at least 0.5 d). Our implementation matches `wotan` to a median of 0.8 ppm.

3.2. *Periodic search*

The periodic search is a box least-squares search (BLS; G. Kovács et al. 2002) over 0.5–100 d. The period range is split into 13 sub-ranges, each testing durations from 0.5 h to a quarter of its shortest period, so that short-period, long-duration transits are not aliased to twice their period. For every trial the depth significance is computed with the noise of the in-transit mean measured at the duration's own timescale, which includes correlated noise as in the combined differential photometric precision of the SPOC Transiting Planet Search (J. M. Jenkins et al. 2010); a detection requires SNR ≥ 7.1. A detection is discarded when it does not survive the removal of its strongest event (the signal should retain SNR $\sqrt{(N-1)/N}$ of its N events within a factor of two), and when it rests on events within 0.5 d of data gaps. Periods $P/2$ and $P/3$ are preferred when folding there is stronger and consistent. Up to three signals per star are reported; each is masked before searching for the next.

On 300 held-out injections of trapezoidal transits into single sectors of planet-host stars, at the same false-alarm rate measured on inverted light curves, the search recovers 74%, 79% and 84% of the detectable signals at false-alarm rates of 1%, 5% and 10%, compared to 69%, 77% and 77% for `wotan` combined with Transit Least Squares (TLS; M. Hippke & R. Heller 2019); the differences are within the bootstrap uncertainties, and TLS recovers more signals at SNR 7–10.

3.3. *Long-period search*

Depending on the observing baseline and gaps, long-period planets may produce only one or a few observed transits. After masking the periodic signals and every catalogued transit, the light curve is detrended with windows of three durations and searched with a matched box filter at every time step for durations of 2–25 h, with the correlated noise of each sector; events with single-event SNR ≥ 8 are refined in time and duration. Each event is vetted against data gaps (fail within 0.5 d), asymmetry between its two halves (fail above 4σ and 30% of the depth), a V-shaped profile, the CCD common mode (Section 3.6), a centroid test with an empirical significance (Section 3.5), and a per-star noise ceiling: the same search on the star's inverted light curve gives the strongest dip that its own noise produces, and the event must exceed it by 30%. Two events with depths within 3σ and durations within a factor of 1.5 form a duo, whose period aliases $\Delta t/n$ are pruned when a predicted transit falls on covered data without a dip. On the inverted

light curves of 100 stars with at least 30 sectors, about one noise event per 100 stars survives this vetting. For single transits with periods of 150–1000 d injected into the same light curves, by expected SNR: 8–12: 45% detected and 9% passing (11 injections); 12–20: 75% detected and 50% passing (16 injections); 20–40: 77% detected and 46% passing (13 injections); > 40: 100% detected and 80% passing (25 injections). The vetting of isolated events is therefore conservative: only 9–50% of injections in the tested SNR bins below 40 pass the full selection. The small numbers in each bin limit the precision of these recovery estimates. The inverted-light-curve result is an empirical noise check, not a measurement of the astrophysical false-positive rate.

3.4. *Photometric vetting*

Every periodic signal is phase-folded with a transit-protected detrending and tested for: SNR with event-to-event scatter (fail below 7.1); odd–even depth difference (fail above 4σ, requiring two events per parity); a secondary eclipse at any phase (Bonferroni-corrected threshold, failing above 5σ when the implied albedo exceeds 1); a radius above $25\,R_\oplus$; a transit duration incompatible with the TIC stellar density; a V-shaped trapezoid fit; crowding (SPOC CROWDSAP); and the centroid shift.

3.5. *Centroids*

For periodic signals, the in-transit shift of the flux-weighted centroid is converted into the position of the source that would produce the observed depth, $\Delta c = \delta(c - x_s)/(1 - \delta)$, and compared with the target position from the world coordinate system of the SPOC aperture, with a 0.1-pixel systematic floor; a source more than 4σ and $21''$ away fails. For single and duo events, whose shifts are not averaged over many transits, the significance is empirical: the shift is ranked among the shifts of the same window at every other time of all sectors, with a generalized Pareto tail beyond the sample.

3.6. *Systematics shared by stars on a CCD*

For each sector, camera and CCD, the median normalized flux of all downloaded stars (2457 stars, catalogued transits masked) forms a common-mode light curve in 30-minute bins. A dip in the common mode at the times of a signal's transits, above 4σ of its correlated scatter, fails the signal as instrumental.

3.7. *Neural-network classifier*

The classifier is a one-dimensional convolutional network with four views of the folded light curve (global, local, odd/even and secondary phase), each processed by a residual branch, plus 14 scalar features from the vetting (depth, SNR, odd–even and secondary significance, ingress fraction, duration ratio, number of transits, centroid significance, crowding, stellar radius, $\log g$, $T_{\rm eff}$, TESS magnitude and density ratio); the network does not see identifiers, dispositions or catalogue planet parameters, and the catalogue ephemeris is only used to align the views. Each labelled TOI contributes one example with all sectors and, for training only, one example per sector. Training uses AdamW with class weighting and early stopping on a validation set.

Performance is estimated with five outer folds grouped by star. Within each outer training set, one of four stratified, star-grouped splits provides the validation set for early stopping and threshold selection. Each star is in exactly one outer test set, scored after the model weights and threshold are fixed. The validation threshold maximizes recall subject to empirical precision of at least 95% and at least five selected examples. Training uses a maximum of 100 epochs, batches of 64 examples and an early-stopping patience of 15 epochs. Table 1 reports the held-out predictions with these fold-specific thresholds; the measured precision is 94.4%.

For the search, we average the five model scores and apply a final threshold selected from the pooled out-of-fold predictions. An isotonic regression fitted to those predictions maps raw scores to calibrated scores. The final threshold and calibration are not tested on an independent set, so the calibrated value is a classifier score rather than a planet probability. Because the calibration map is a step function, selected and unselected signals can share the same calibrated score.

3.8. *Catalogue cross-match*

A signal is considered catalogued when it matches, by period (or a harmonic) and phase, a TOI, a community TOI (ExoFOP), a planet of the NASA Exoplanet Archive with a TIC identifier, a SPOC threshold-crossing event (TCE) from any of the 132 single- and multi-sector data validation runs listed on MAST (341 351 TCEs), or an eclipsing

Table 1. Cross-validated performance of the classifier.

Quantity	Value
Labelled TOIs (stars)	2277 (2134)
Positives (CP/KP)	1349
Average precision	0.932
Average precision, per fold	0.933 ± 0.015
ROC AUC	0.919
Precision at threshold	0.944 (0.924–0.958)
Recall at threshold	0.508
Confusion (TP, FP, FN, TN)	685, 41, 664, 887

NOTE—Out-of-fold scores of five models, folds grouped by star; fold-specific validation thresholds target 95% precision. The precision interval is the Wilson 95% interval.

binary of A. Prša et al. (2022). The phase tolerance grows with the propagated 3σ ephemeris drift; once it exceeds a quarter of the period, the period alone decides.

3.9. *Difference images*

For signals that pass the vetting and the classifier, the SPOC target pixel files of up to three sectors with transits are downloaded and the mean out-of-transit image minus the mean in-transit image is fitted with a Gaussian. Its offset from the target position (same stamp, same WCS, 0.1-pixel floor) fails the signal above 4σ and 0.5 pixel ($\sim 10''$). On TFOP dispositions this test failed 7 of 13 false positives labelled as nearby eclipsing binaries and 0 of 5 confirmed planets with a usable difference image.

3.10. *False-positive probabilities*

`TRICERATOPS` (S. Giacalone et al. 2021, 2022) computes the false-positive probability (FPP) and the nearby false-positive probability (NFPP) from the folded transit, the TIC and Gaia neighbours and the SPOC apertures. We provide the pixel geometry from the aperture extension of the local light-curve files instead of TESScut image cutouts; the relative pixel positions agree with TESScut to 0.001 pixel. The FPP is a Monte Carlo estimate whose run-to-run scatter is significant: the search averages three runs, and the candidate analysis averages ten. S. Giacalone et al. (2021) adopt FPP $< 1.5\%$ and NFPP $< 0.1\%$ as validation criteria. We use the more conservative FPP $< 1\%$ threshold with the same NFPP limit. Both probabilities depend on the scenario priors and on the available constraints; without high-resolution imaging, close companions are constrained only by the TIC and Gaia.

3.11. *Tiers*

A signal is *rejected* when a vetting test or the pixel test fails or when NFPP $> 10\%$; *inconclusive* when it passes the vetting but not the classifier threshold, or passes both but has FPP $\geq 50\%$ or $0.1\% \leq$ NFPP $\leq 10\%$; a *candidate* when it passes both and has FPP $< 50\%$ and NFPP $< 0.1\%$; and *strong* when a candidate also has FPP $< 1\%$. A difference image that is not significant neither passes nor fails the pixel test. In the search, difference images and `TRICERATOPS` are only computed for signals selected by the classifier; we also computed them for the inconclusive signals of the open sample (Section 4.3). Table 2 gives the tiers of the search, before these additional tests.

3.12. *Transit fits*

The four candidates are fitted with a quadratic limb-darkened model (K. Mandel & E. Agol 2002) with the q_1, q_2 parameterization (D. M. Kipping 2013), free mid-transit time, period, radius ratio, impact parameter, stellar density, flux offset and white-noise amplitude, assuming circular orbits. Priors are uniform in the sampled coordinates: $R_p/R_\star \in (0.001, 0.5)$, $b \in (0, 1.5)$, $\log_{10}[\rho_\star/(\mathrm{g\,cm^{-3}})] \in (-3, 2)$, $q_1, q_2 \in (0, 1)$, relative flux offset $f_0 \in (-0.01, 0.01)$, and

Table 2. Signals found in the 132 host stars of open TOIs.

Search	Tier	Catalogued	Not catalogued
periodic	candidate	4	0
periodic	inconclusive	94	15
periodic	rejected	17	42
single	rejected	3	26

NOTE—Tiers refer to the initial search, before supplementary testing. Catalogued signals match a TOI, CTOI, published planet, SPOC TCE or known eclipsing binary; these counts are not restricted to the open TOIs.

$\ln\sigma \in (-14, -2)$ for relative-flux noise amplitude σ. The mid-transit time is restricted to half an initial transit duration on either side of its initial estimate, and the period to $\pm 0.1\%$ of its initial estimate. We require $b \leq 1 + R_p/R_\star$ and $a/R_\star > 1 + R_p/R_\star$. The density and noise priors are therefore log-uniform, not uniform in linear units. The model averages 15 instants per 120 s exposure; its instantaneous flux matches `batman` (L. Kreidberg 2015) to 10^{-7}. Errors are scaled by the time-averaging factor β for correlated noise (J. N. Winn et al. 2008), and the posterior is sampled with `emcee` (D. Foreman-Mackey et al. 2013) using differential-evolution moves until the chains exceed 50 autocorrelation times. Planet radii use the TIC v8 stellar radii and their uncertainties.

4. RESULTS

4.1. *Recovery of the open TOIs*

Table 2 summarizes the outcome. The blind search recovers 110 (83%) of the 133 open TOIs. Of the 23 TOIs not recovered, 1 has a period below the 0.5 d limit of the periodic search, 6 are single transits or have periods beyond 100 d (their events were found by the long-period search but rejected by its conservative vetting or not detected), 4 have a catalogue SNR below 10, and 12 have higher SNR: for these the 120 s sectors available do not always include the sectors in which the TOI was detected.

4.2. *Candidates*

Four recovered TOIs meet the search candidate-selection criteria (Table 3, Figure 1). The difference images of TOI-3025.01, TOI-4674.01 and TOI-7850.01 place the transit source 1.3–2.9″ from the target, within 1.2σ (Figure 2); TOI-7888.01 has a single 120 s sector with transits and its difference image is not significant (SNR 4.2), so its source is only constrained by the centroid test and `TRICERATOPS`. All four have NFPP $\leq 2 \times 10^{-7}$, so nearby eclipsing binaries are unlikely, but their FPPs (1.7–34%) are dominated by eclipsing-binary and hierarchical scenarios on the target itself: with radii of 6.7–13.9 $R_\oplus$ the transit depth alone cannot distinguish a giant planet from a brown dwarf or a low-mass star. TOI-3025.01 (FPP $= 1.7 \pm 0.4\%$) is the closest to the validation threshold.

Table 3. Candidates: transit fits, classifier score and false-positive probabilities.

TOI	TIC	$T_0 - 2460000$	P (d)	$R_p/R_\star$	b	T_{14} (h)	R_p ($R_\oplus$)	$T_{\rm eq}$ (K)	CNN	FPP (%)	NFPP
TOI-3025.01	452464529	$92.5253^{+0.0007}_{-0.0007}$	$3.67950^{+0.00001}_{-0.00001}$	$0.0896^{+0.0011}_{-0.0011}$	$0.18^{+0.15}_{-0.12}$	$4.28^{+0.05}_{-0.05}$	$13.9^{+1.0}_{-1.0}$	1553^{+43}_{-38}	0.92	1.7 ± 0.4	6×10^{-11}
TOI-4674.01	89062142	$1240.9832^{+0.0010}_{-0.0010}$	$4.24307^{+0.00017}_{-0.00017}$	$0.0807^{+0.0022}_{-0.0022}$	$0.40^{+0.21}_{-0.25}$	$3.87^{+0.10}_{-0.09}$	$13.5^{+1.0}_{-1.0}$	1365^{+94}_{-59}	0.92	33.5 ± 0.5	7×10^{-9}
TOI-7850.01	410056850	$1157.3067^{+0.0010}_{-0.0010}$	$10.60528^{+0.00034}_{-0.00033}$	$0.0848^{+0.0020}_{-0.0028}$	$0.56^{+0.11}_{-0.24}$	$5.25^{+0.11}_{-0.11}$	$12.4^{+0.7}_{-0.7}$	1217^{+66}_{-69}	0.92	7.1 ± 0.6	1×10^{-7}
TOI-7888.01	219396153	$668.7548^{+0.0036}_{-0.0035}$	$10.23700^{+0.00378}_{-0.00336}$	$0.0950^{+0.0096}_{-0.0074}$	$0.49^{+0.28}_{-0.32}$	$3.32^{+0.33}_{-0.25}$	$6.7^{+0.7}_{-0.6}$	572^{+84}_{-48}	0.93	15.5 ± 0.7	0

NOTE—T_0 in $\mathrm{BJD_{TDB}}$; $T_{\rm eq}$ for zero albedo and full redistribution. CNN is the calibrated classifier score. FPP and NFPP from ten TRICERATOPS runs (mean and error of the mean), without high-resolution imaging. Three significant difference images are consistent with the target position; the image for TOI-7888.01 is not significant. Zero NFPP values reflect numerical estimates, not an exclusion of all contaminating sources.

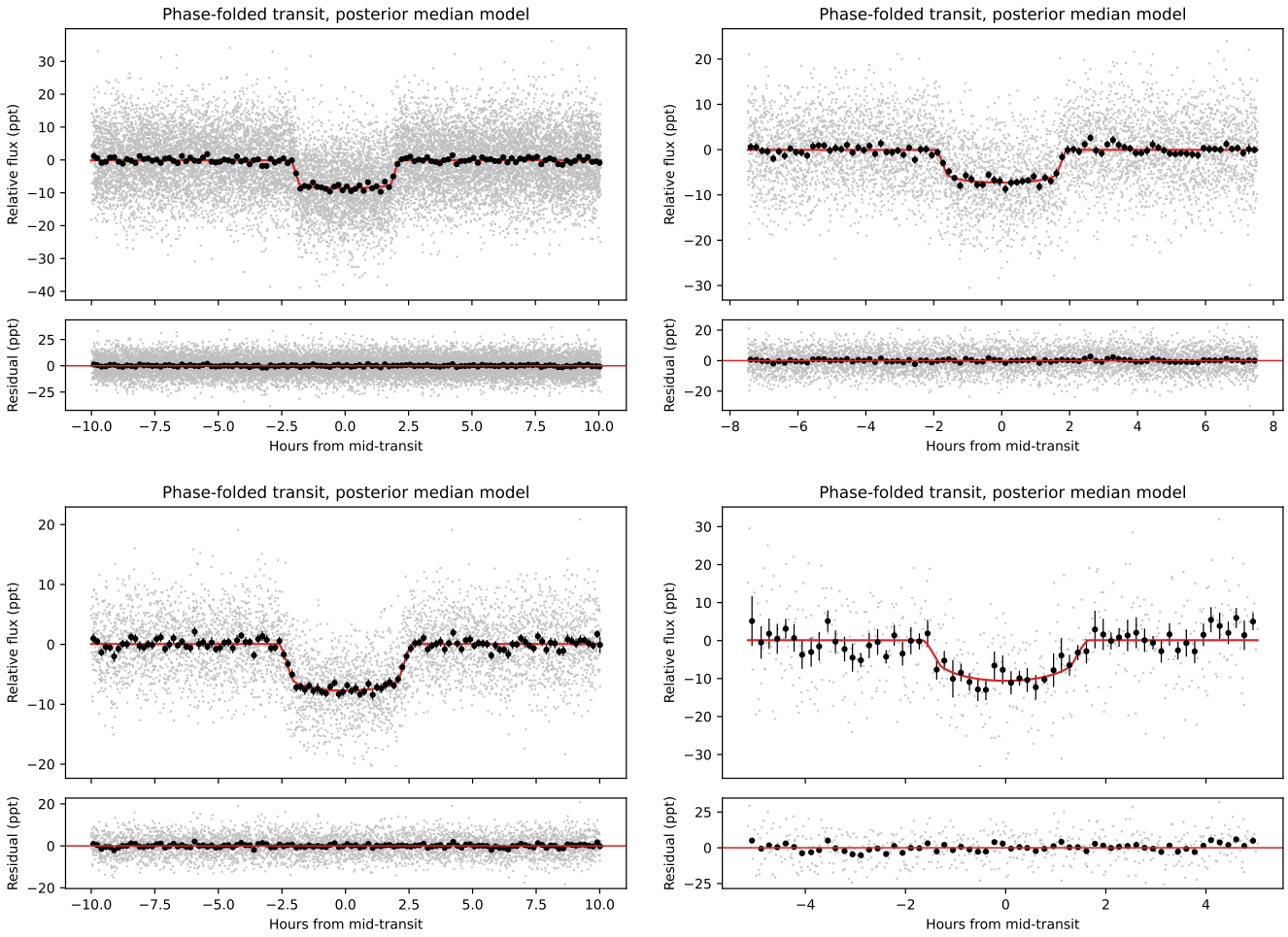


Figure 1. Phase-folded transits of TOI-3025.01 (top left), TOI-4674.01 (top right), TOI-7850.01 (bottom left) and TOI-7888.01 (bottom right) with the posterior median model; black points are 10-minute bins.

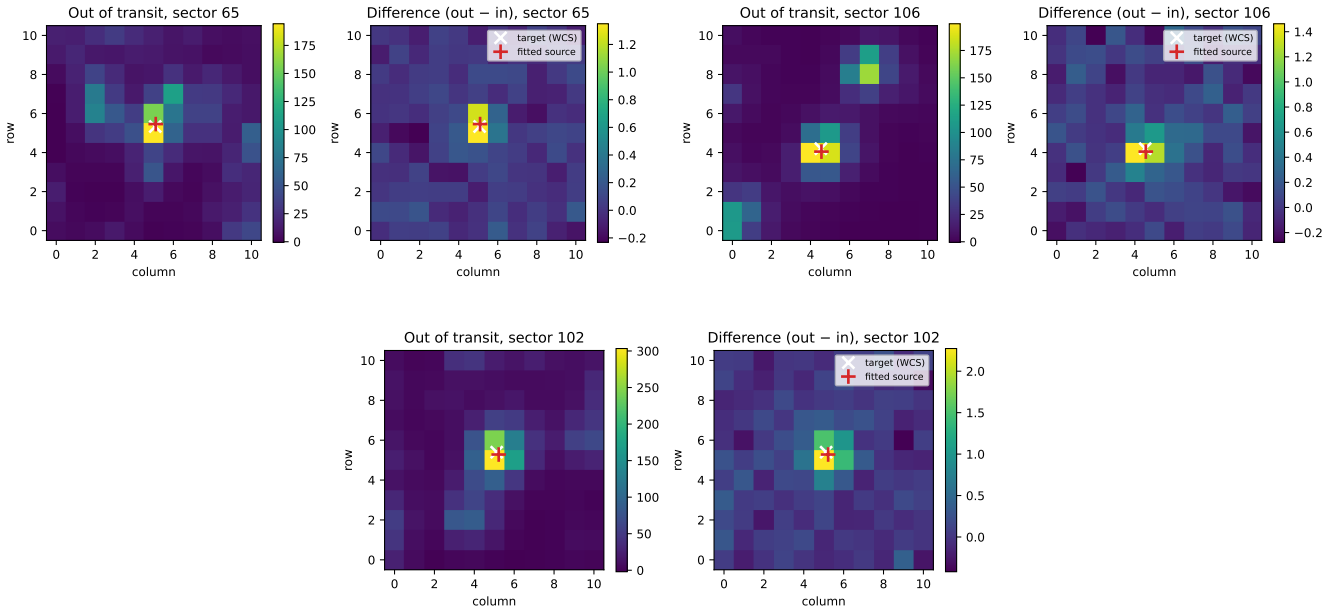


Figure 2. Out-of-transit and difference images of TOI-3025.01 (top left), TOI-4674.01 (top right) and TOI-7850.01 (bottom) in one sector each (the difference image of TOI-7888.01 is not significant). White crosses mark the catalogue position of the target; red plus signs mark the fitted source of the transit signal.

4.3. *Supplementary tests of initially inconclusive TOIs*

A total of 91 recovered TOIs passed the photometric vetting and were initially classified as inconclusive. Of these, 86 score below the classifier threshold (raw scores 0.02–0.876, threshold 0.877). The other 5 (marked with † in Table 4) are selected by the classifier, but the search `TRICERATOPS` result falls outside the candidate criterion (FPP $\geq$ 50%, or 0.1% $\leq$ NFPP $\leq$ 10%). Difference images and `TRICERATOPS` (three runs) were computed for all of them. None has FPP $<$ 1% and NFPP $<$ 0.1%, while 17 have FPP $<$ 50% and NFPP $<$ 0.1%, 49 have NFPP $>$ 10% (signal likely from a neighbour), and 2 show a significant difference-image offset ($> 4\sigma$ and > 0.5 pixel). Signals above the NFPP or offset limits would be rejected by these tests; they are kept in the table to show the full group.

Table 4. Initially inconclusive recovered TOIs: supplementary source tests and false-positive probabilities.

TOI	TIC	P (d)	Depth (ppm)	SNR	R_p ($R_\oplus$)	CNN	FPP	NFPP	Offset ($''$)	Offset (σ)
TOI-2542.01	431971818	4.1099	33243	13.4	–	0.46	0.923	0.62	–	–
TOI-2861.01	157572174	6.6301	5685	14.1	12.3	0.46	0.132	0.023	3.2	0.9
TOI-2897.01†	308987429	12.0041	10994	40.8	11.2	0.92	1.000	1×10^{-4}	0.7	0.1
TOI-2926.01	34499178	2.4249	7235	18.1	17.8	0.74	0.063	0.013	3.8	2.0
TOI-2993.01	48768158	2.3372	5934	18.2	–	0.85	0.800	0	1.5	0.4
TOI-3012.01	437560683	3.0919	61168	29.8	–	0.46	0.875	0.38	0.4	0.1
TOI-3014.01	383806208	1.7947	3859	25.2	11.4	0.18	1.000	0.98	0.5	0.1
TOI-3021.01	45054849	11.2598	17883	30.9	–	0.85	0.889	0.44	0.5	0.1
TOI-3034.01	391645374	1.4033	18905	40.4	–	0.05	0.967	0.83	–	–
TOI-3043.01	360706000	2.3726	6797	24.2	18.7	0.18	0.994	0.6	1.2	0.4
TOI-3062.01	463343283	4.8650	10389	15.4	13.5	0.40	0.994	0.98	6.1	3.2
TOI-3074.01	447976172	5.3583	9486	26.6	–	0.85	0.933	0.67	3.6	1.4
TOI-3079.01	441444262	1.9898	20273	64.4	–	0.46	0.938	0.69	1.1	0.4
TOI-3081.01	432467894	3.4364	6942	23.2	23.9	0.68	1.000	0.045	0.9	0.2
TOI-3090.01	309913305	6.3273	9067	10.3	–	0.66	0.957	0.78	8.5	4.8
TOI-3143.01	384244834	2.9903	20475	37.5	–	0.18	0.996	0.97	3.2	2.0
TOI-3172.01	398502253	3.0041	6468	11.9	16.8	0.42	0.529	0.29	2.7	0.9
TOI-3208.01	319949055	2.3577	12193	13.4	16.8	0.42	0.065	5×10^{-4}	1.4	0.4
TOI-3209.01	319806044	1.3509	9002	9.4	–	0.35	1.000	0.99	2.0	0.6
TOI-3258.01	369789627	1.2280	2695	20.7	–	0.46	0.929	0.64	5.7	3.1
TOI-3307.01	369861728	3.8781	9744	21.6	12.3	0.42	0.495	0.4	–	–
TOI-3387.01	118072143	2.7678	10350	22.0	–	0.18	0.923	0.62	1.8	0.7
TOI-3393.01	173260345	5.2060	12316	19.0	13.7	0.68	0.546	0.34	1.4	0.5
TOI-3421.01	145023816	2.8389	6912	19.3	19.4	0.35	0.458	0.08	0.9	0.1
TOI-3425.01	144682439	0.7654	3300	12.0	9.5	0.29	0.974	0.78	–	–
TOI-3439.01	394338713	5.9350	13543	32.1	14.4	0.18	0.979	0.56	1.9	0.8
TOI-3463.01	292889298	2.2532	19798	17.2	–	0.75	0.944	0.72	–	–
TOI-4169.01	372747949	3.4027	3766	45.8	14.6	0.75	0.278	2×10^{-6}	1.7	0.6
TOI-4228.01	415143674	6.0167	13460	19.1	20.1	0.46	0.354	0.13	1.3	0.2
TOI-4720.01	124322274	3.7814	11474	12.2	18.5	0.46	0.356	0.015	0.4	0.0
TOI-4793.01	65949719	3.4629	11027	24.9	17.1	0.68	0.177	8×10^{-5}	1.0	0.1
TOI-4796.01	65905621	3.5776	10545	27.9	18.1	0.42	0.716	0.36	1.0	0.2

Table 4 *continued on next page*

Table 4 *(continued)*

TOI	TIC	P (d)	Depth (ppm)	SNR	R_p ($R_\oplus$)	CNN	FPP	NFPP	Offset ($''$)	Offset (σ)
TOI-4802.01	50575306	1.7167	8496	15.2	10.9	0.08	0.995	0.99	12.8	5.0
TOI-4815.01	19447218	8.9727	10405	11.1	13.5	0.74	0.484	0.3	4.4	1.2
TOI-4870.01	446135868	1.8119	7871	19.7	9.3	0.35	0.992	0.98	5.6	2.9
TOI-4874.01	463754499	2.2867	4967	11.3	18.6	0.35	0.866	0.47	8.0	2.4
TOI-4881.01	359735804	4.4159	8141	28.7	19.9	0.68	0.238	4×10^{-4}	0.2	0.0
TOI-4884.01	80681136	1.5305	5133	18.3	–	0.42	0.941	0.71	1.6	0.3
TOI-4894.01	469333534	3.5611	7434	8.8	–	0.42	0.957	0.78	10.2	3.9
TOI-4900.01	425508314	7.6650	8481	28.2	16.9	0.85	0.602	5×10^{-11}	–	–
TOI-4910.01	303327435	5.5075	16331	15.8	14.1	0.50	0.785	0.3	–	–
TOI-4924.01	307099276	3.1468	14486	18.4	–	0.35	0.994	0.83	1.5	0.2
TOI-4943.01	418567812	8.0176	19106	21.9	–	0.85	0.329	0.31	0.4	0.1
TOI-4975.01	304085024	4.4835	7228	19.8	–	0.19	0.999	0.87	–	–
TOI-5033.01[†]	318180448	5.4696	1506	12.2	1.5	0.96	0.090	4×10^{-3}	–	–
TOI-5039.01	396649769	2.1968	5442	8.1	14.5	0.66	0.506	0.037	3.4	1.1
TOI-5046.01	306512967	11.1837	5939	17.5	–	0.75	0.206	0.18	–	–
TOI-5064.01	407127579	3.5108	1223	16.7	6.1	0.66	0.273	0.23	–	–
TOI-5922.01	364192929	5.2054	11149	17.6	20.6	0.66	0.818	0	4.4	1.0
TOI-6259.01	294497466	3.2387	559	7.1	5.6	0.05	0.810	0.22	–	–
TOI-6453.01	66242670	6.8669	5412	16.8	15.9	0.42	0.403	0.37	10.0	2.8
TOI-6464.01	190643105	9.6741	3653	23.6	17.5	0.68	0.014	6×10^{-6}	3.7	0.9
TOI-6471.01	410901394	3.0761	2399	10.8	6.1	0.66	0.485	0.43	–	–
TOI-6472.01	140706833	2.6819	6294	20.8	17.3	0.42	0.270	0.26	2.6	0.8
TOI-6477.01	11927677	3.6861	7626	46.3	16.8	0.42	0.221	0.02	2.1	0.5
TOI-6485.01	281286325	17.0184	14804	32.0	16.7	0.89	0.456	0	2.9	1.3
TOI-6487.01	37754672	7.0678	3754	19.8	10.3	0.42	0.659	0.017	2.3	0.6
TOI-6490.01	118754987	7.8420	7118	19.9	14.1	0.75	0.062	6×10^{-7}	5.3	1.0
TOI-6504.01	177257073	6.8327	5072	25.3	11.3	0.75	0.792	0.78	1.1	0.1
TOI-6536.01	280647591	4.2379	6311	13.8	8.5	0.75	0.953	0.94	0.2	0.0
TOI-6537.01	80384677	3.6480	3001	10.6	–	0.74	0.787	0.75	–	–
TOI-6544.01	161738745	3.6250	874	18.2	–	0.85	0.933	0.67	–	–
TOI-6545.01	72504017	5.6178	450	8.9	5.7	0.18	0.598	0.056	–	–
TOI-6554.01	94483093	5.2309	1100	12.6	5.3	0.46	0.209	0.16	–	–
TOI-6555.01	259606227	10.1286	526	12.9	4.7	0.85	0.009	2×10^{-3}	–	–
TOI-6558.01	420529106	9.5307	1399	14.9	2.9	0.45	0.019	4×10^{-3}	–	–
TOI-6565.01	392914193	0.8775	1960	21.7	7.1	0.42	0.482	9×10^{-13}	2.6	0.5
TOI-6590.01	72123144	7.0838	8442	9.2	13.7	0.43	0.543	0.48	6.2	1.3
TOI-6591.01	463868678	5.7831	1504	12.1	15.8	0.42	0.291	7×10^{-4}	–	–
TOI-6595.01	361344753	7.5157	1454	13.7	3.7	0.89	0.079	0.05	–	–
TOI-6827.01	302453291	4.4642	2690	14.4	14.3	0.85	0.472	2×10^{-4}	–	–
TOI-7465.01	167804011	6.2057	4552	19.7	8.7	0.89	0.012	0	3.3	0.7
TOI-7488.01	128168083	3.9050	19046	17.4	17.6	0.85	0.539	1×10^{-3}	4.1	0.9

Table 4 *continued on next page*

Table 4 *(continued)*

TOI	TIC	P (d)	Depth (ppm)	SNR	R_p ($R_\oplus$)	CNN	FPP	NFPP	Offset ($''$)	Offset (σ)
TOI-7602.01	156716001	10.3048	656	8.0	3.0	0.86	0.017	0.0074	–	–
TOI-7639.01†	270471727	10.4564	662	11.7	2.8	0.92	0.009	1×10^{-3}	–	–
TOI-7682.01	159671133	1.3438	168	8.4	2.8	0.18	0.619	0.51	–	–
TOI-7684.01	115942852	3.2112	7992	9.0	6.6	0.86	0.359	0.3	0.1	0.0
TOI-7686.01	273861354	13.9967	15774	13.2	9.2	0.86	0.392	0.06	3.1	0.4
TOI-7689.01	12999193	1.4830	20865	25.3	4.0	0.85	0.954	0	5.5	1.9
TOI-7701.01	122522333	20.6077	2053	29.8	8.7	0.85	0.038	8×10^{-6}	92.7	8.0
TOI-7705.01	70514245	4.0478	20686	80.0	12.6	0.86	0.306	3×10^{-4}	0.7	0.2
TOI-7711.01	1309019	3.2423	9027	19.4	13.8	0.46	0.109	0.0098	3.4	0.9
TOI-7712.01	264460009	3.0934	22301	25.1	15.3	0.46	0.902	0.85	0.6	0.0
TOI-7713.01	77287067	4.0763	14294	30.0	16.9	0.85	0.052	4×10^{-5}	1.5	0.4
TOI-7720.01	421184129	8.0315	1807	13.8	14.7	0.18	0.283	3×10^{-7}	–	–
TOI-7723.01†	162093661	6.3338	5631	7.3	4.6	0.92	0.039	0.02	–	–
TOI-7762.01	193318850	7.8592	4920	12.7	6.1	0.89	0.102	0	–	–
TOI-7822.01	146320244	4.4938	1820	11.9	5.0	0.75	0.257	0.2	–	–
TOI-7824.01†	73644471	4.5966	9726	36.7	15.4	0.93	0.586	0	6.2	1.9
TOI-7825.01	193096383	4.8565	3531	18.2	6.3	0.74	0.043	0	1.3	0.1
TOI-7854.01	124095894	1.1257	9926	12.0	–	0.42	0.909	0.55	–	–

NOTE—Depth, SNR and radius from the box model of the vetting; FPP and NFPP from three TRICERATOPS runs; offset of the difference-image source from the target. Dashes: not computed or image not significant. Membership in this table follows the initial search tier; supplementary rejection criteria are discussed in the text.

The supplementary tests separate a few cases. TOI-4802.01 ($P = 1.72$ d) is most likely a nearby eclipsing binary: the difference-image source lies $12.8''$ from the target at 5σ and NFPP = 0.99. TOI-7701.01 ($P = 20.6$ d) has a low FPP (0.04), but its only significant difference image (one transit, sector 106) places the source $93''$ (4.4 pixels) from the target at 8σ; the FPP does not account for this, and the signal should be treated as a possible blend until more transits are imaged. Both fail the pixel test. Among the signals located on the target, TOI-7465.01, TOI-6464.01, TOI-7825.01 and TOI-7713.01 have FPP $\approx$ 1–5% and NFPP $< 0.01\%$ from three runs. TOI-7465.01, with FPP = 1.2%, has a raw classifier score of 0.856, just below the 0.877 threshold. These four are worth following up even though the classifier did not select them.

4.4. *Signals not in any catalogue*

A total of 15 signals in these stars pass the photometric vetting but match no catalogued signal (Table 5). None is selected by the classifier. Each lies in the host star of an open TOI and does not coincide with it in period or phase; they are listed with the same supplementary tests as a starting point for inspection, not as planet candidates.

Table 5. Signals that pass the photometric vetting and match no catalogued signal.

TIC	Host TOI	P (d)	$T_0 - 2460000$	T_{14} (h)	Depth (ppm)	SNR	$N_{\rm tr}$	R_p ($R_\oplus$)	CNN	FPP	NFPP	Offset ($''$)
66620917	TOI-4391.01	0.6166	69.3666	0.75	1470	15.8	102	–	0.42	1.000	8×10^{-8}	17.1
81089255	TOI-6480.01	66.5924	-744.7652	1.50	2571	8.2	2	7.9	0.40	0.880	0.26	93.6
105712339	TOI-7754.01	22.9961	1067.4995	1.50	6892	14.2	2	10.1	0.18	0.786	0.096	–
255560319	TOI-6703.01	92.6536	-1605.7942	1.50	448	7.4	3	5.7	0.35	1.000	0	–

Table 5 *continued on next page*

Table 5 *(continued)*

TIC	Host TOI	P (d)	$T_0 - 2460000$	T_{14} (h)	Depth (ppm)	SNR	$N_{\rm tr}$	R_p ($R_\oplus$)	CNN	FPP	NFPP	Offset ($''$)
259606227	TOI-6555.01	1.0644	1046.5423	2.50	185	12.1	60	2.8	0.08	0.066	0.061	–
261746870	TOI-7710.01	72.0614	-1658.7563	0.50	2550	12.6	2	5.0	0.46	0.999	0	–
292889298	TOI-3463.01	0.5014	69.0558	1.00	9246	8.8	102	–	0.35	0.973	0.86	–
294497466	TOI-6259.01	0.6550	718.6830	2.00	267	7.8	79	3.9	0.01	0.134	0.12	–
303327435	TOI-4910.01	61.1731	34.0786	10.00	14113	9.6	3	13.1	0.42	0.603	0.52	–
306512967	TOI-5046.01	0.7532	-35.2373	2.50	481	7.3	479	–	0.04	0.342	0.33	–
369861728	TOI-3307.01	13.0871	69.6090	6.00	3324	8.5	3	7.2	0.46	0.628	0.62	–
391645374	TOI-3034.01	1.4908	15.6886	2.50	3165	7.6	40	–	0.19	0.989	0.95	–
407827885	TOI-4963.01	41.7279	99.4532	0.75	14298	7.7	2	–	0.85	0.955	0.77	–
420529106	TOI-6558.01	21.9609	1049.7166	3.00	954	9.6	4	2.4	0.85	0.034	0.013	–
425508314	TOI-4900.01	13.0538	54.8459	1.50	4741	10.2	4	12.6	0.35	0.791	0.095	–

NOTE—Box-model parameters from the vetting; T_0 in $\mathrm{BJD_{TDB}}$. FPP, NFPP and difference-image offset from the supplementary analysis (three TRICERATOPS runs).

Two of these signals are off target: the difference images of TIC 66620917 ($P = 0.62$ d) and TIC 81089255 ($P = 66.6$ d) place the source $17''$ (12σ) and $94''$ (26σ) from the target, so both are attributed to neighbouring stars. The most promising is TIC 420529106 ($P = 21.96$ d, four transits, depth 954 ppm, $R_p \approx 2.4\,R_\oplus$, SNR 9.6), with a calibrated score of 0.85, FPP = 0.03 and NFPP = 0.01; its difference image is not significant. TIC 259606227 ($P = 1.06$ d, $R_p \approx 2.8\,R_\oplus$) has FPP = 0.07 but a classifier score of 0.08. At SNR near 10, both need more data before they can be considered candidates.

5. DISCUSSION

Classifier domain. The network is trained on labelled TOIs, and its precision applies to that population. On 200 inverted single-sector light curves the full pipeline selected one noise signal, but the classifier was designed for high precision and selects about half of the confirmed planets; a classifier score below threshold is weak evidence against a planet.

False-positive probabilities. The FPPs are computed without high-resolution imaging, so close bound companions are only constrained by the TIC and Gaia. The Monte Carlo scatter of `TRICERATOPS` is comparable to the differences between some candidates, which is why the candidate values average ten runs; the quoted errors cover only this scatter, not the priors. An NFPP of zero means that no simulated nearby scenario reproduced the transit, not that contamination is excluded. None of our signals is statistically validated.

Completeness. The sample is restricted to stars with 120 s data. The periodic search covers 0.5–100 d, and the isolated-event search has limited sensitivity to longer periods. TOIs detected in sectors without 120 s data may be missed. The recovery fraction of catalogued TOIs is not a population completeness, and the injection tests use trapezoidal signals.

Follow-up. The candidates are suitable for reconnaissance spectroscopy, which would identify eclipsing binaries through radial-velocity variations of kilometres per second, and for high-resolution imaging, which would constrain close companions and inform revised FPPs. In particular, TOI-3025.01 (FPP $= 1.7 \pm 0.4\%$) is close to the statistical validation threshold and represents the highest-priority target among the four candidates.

6. CONCLUSIONS

A pipeline that combines a blind periodic and long-period search, photometric, centroid, systematic and pixel-level tests, a cross-validated neural network and `TRICERATOPS` recovers 110 (83%) of the 133 open TOIs with 120 s data. Four of them meet our candidate-selection criteria, three have difference images consistent with the target position, and their FPPs are 1.7–34%. We list the results for every recovered TOI and the uncatalogued signals that pass the photometric vetting, as a prioritization aid for follow-up observations: after the four candidates, the on-target TOIs TOI-7465.01, TOI-6464.01, TOI-7825.01 and TOI-7713.01 and the uncatalogued signal of TIC 420529106 are the most promising.

DATA AND CODE AVAILABILITY

The TESS observations used in this work are available from MAST; the TOI catalogue and follow-up records were retrieved from ExoFOP as described in Section 2. The analysis code and results are available at https://github.com/nrolabs/tess-open-tois.

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Facilities: TESS, MAST, ExoFOP

Software: `astropy` (Astropy Collaboration 2022), `batman` (L. Kreidberg 2015), `emcee` (D. Foreman-Mackey et al. 2013), `lightkurve` (Lightkurve Collaboration 2018), `numpy` (C. R. Harris et al. 2020), `PyTorch` (A. Paszke et al. 2019), `scikit-learn` (F. Pedregosa et al. 2011), `scipy` (P. Virtanen et al. 2020), `TRICERATOPS` (S. Giacalone et al. 2021), `pytransit` (H. Parviainen 2015)

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